heyyERP for V-Rub — Business Requirements Document (BRD)

Oct 22, 2026 · @Heyybot

1. Document control and scope

This BRD is the build specification for heyyERP at V-Rub Suspensions Pvt. Ltd. It turns the approved clickable prototype (v4) into requirements, code structure and target architecture that the HeyyBot engineering team can estimate, build and test against.

Field	Value
Client	V-Rub Suspensions Pvt. Ltd. (automotive anti-vibration rubber and metal-bonded parts)
Product	heyyERP — cloud, multi-plant, multi-GSTIN ERP with HeyyBot AI
Version	1.0 draft, based on prototype build `heyyerp-vrub.html` (state key `heyyerp-vrub-v4`)
Inputs	ERP RFQ; prototype demo feedback; V-Rub workflow template (sheet 7 returned); V-Rub process-flow PDF; PRD v1; quotation HYB-2026-71
Audience	Product, engineering (frontend, backend, data, DevOps), QA, implementation consultants, V-Rub key users
Approvers	HeyyBot: CBO / Founder. V-Rub: MD, Plant Head

In scope: the 11 functional modules in Section 4, the approval engine, role-based access, V-Rub business rules, production routes, migration of masters and opening balances, dashboards and AI copilot.

Out of scope (phase 1): machine / PLC integration, OEM portal integration, payroll processing, historical transaction migration, on-premise hosting.

Requirement ID convention

- `BR-xx` business requirement · `FR-<MOD>-xx` functional requirement · `NFR-xx` non-functional · `RULE-xx` V-Rub business rule.
- Priority: **M** must (go-live), **S** should (within 60 days of go-live), **C** could (phase 2).
- Status of every FR traces to a prototype screen or action (`data-a` action name) so QA can demo-match.

Glossary

Term	Meaning
GRN	Goods Receipt Note; stock lands in QA hold until inspection
FIFO / FEFO	First-in-first-out valuation / first-expiry-first-out issue
Compound batch	Mixed rubber batch; must mature for a minimum time before use and expires after a maximum time
Route	Ordered set of operations and QA gates a part follows (METAL, MOULD, EXTR)
Value band	Amount range in the approval matrix that decides which approvers a document needs
Period lock	Month-end lock after which postings to that month are blocked
Inter-unit invoice	Invoice raised between two V-Rub GSTINs when material moves between units

2. Business context and objectives

V-Rub makes anti-vibration rubber parts for OEMs (suspension bushes, engine mounts, centre-bearing mounts, torque rods) plus EPDM extrusions. Work runs across three plants with separate GSTINs, and today it lives in spreadsheets, an accounting package and paper travellers. heyyERP replaces these with one system of record from vendor approval to dispatch.

Plants and stores (as configured in the prototype)

Unit	Role in the flow	GSTIN (demo placeholder)	Stores
Plant 1 · Moulding & Assembly	Moulding, packing, FG, dispatch	06AAKCV4123R1ZQ	Main Store P1, FG Store P1
Plant 2 · Metal Prep & Bonding	Machining, phosphating, painting of metal inserts	06AAKCV4123R2ZP	Stores P2
Plant 3 · Rubber Mixing	Weighing, mixing, maturation of compound; extrusion	06AAKCV4123R3ZO	Rubber Store P3
Common	Maintenance spares	—	Spares Store

Real GSTINs, plant names and store list come from V-Rub sheet 1 (company setup), not yet returned.

Business requirements

ID	Requirement	Priority
BR-01	One shared database for all three plants with a consolidated and per-unit view	M
BR-02	Every user signs in with their own account; screens and actions follow their role	M
BR-03	Documents route to the right approver by type, value and department, with SLA and audit trail	M
BR-04	Full lot traceability: RM lot → compound batch → metal lot → moulded lot → FG lot → invoice	M
BR-05	Enforce V-Rub's stores rules: FIFO valuation, FEFO issue, no negative stock, monthly count	M
BR-06	Enforce production routes with QA gates and compound maturation windows	M
BR-07	Import OEM schedules from Excel and e-mail into sales orders	M

ID	Requirement	Priority
BR-08	GST-compliant invoicing, inter-unit invoices, e-invoice and e-way bill	M
BR-09	Month-end period lock, with production open until day 1 of the next month	M
BR-10	Role dashboards and an AI copilot that answers questions on live data	S
BR-11	Changes to approval rules and business settings by an admin, without code changes	M
BR-12	Works on shop-floor tablets and phones; barcode labels at every stage	M

Success metrics (targets to agree with V-Rub at BRD sign-off)

- Month-end close in 3 working days or less.
- Stock accuracy of 98% or more at monthly count.
- 100% of moulded lots traceable to compound batch and metal lot.
- Median approval cycle under 24 hours; under 5% of approvals past SLA.
- Zero issues of expired or un-matured compound to the press.

3. Users, roles and permissions

Access is two-layered: a role grants **modules** (what a user can open) and **permissions** (what they can do). Every action button carries an action name, and each action maps to one permission. The backend must enforce the same check on every API call; the UI check alone is not security.

Roles

Role code	Name	Home screen	Modules	Demo user (ID)
ADMIN	System admin	Command center	All	HeyyBot Admin (`admin`)

Role code	Name	Home screen	Modules	Demo user (ID)
MD	Managing Director	Command center	All, all permissions; cannot change admin settings	Pranav Bansal (`md`)
PLANT_HEAD	Plant Head	Command center	All except integrations, go-live, benchmark	Anil Sharma (`plant`)
PURCHASE	Purchase Manager	Command center	Procurement, inventory, routes, reports, kaizen, AI	Kavita Joshi (`purchase`)
STORES	Stores In-charge	Command center	Procurement, inventory, routes, reports, kaizen	Sunil Mehra (`stores`)
QA	Quality Head	Quality	Quality, routes, inventory, manufacturing, procurement, reports	Neha Gupta (`quality`)
PROD	Production Manager	Command center	Routes, manufacturing, inventory, quality, maintenance, reports, AI	Rakesh Yadav (`production`)
SALES	Sales & Dispatch	Command center	Sales, inventory, reports, AI	Ritu Malhotra (`sales`)
ACCOUNTS	Finance Controller	Command center	Finance, GST, procurement, sales, inventory, maintenance, reports, AI	Deepa Iyer (`accounts`)
HR	HR Manager	Command center	HR, kaizen, reports	Shalini Verma (`hr`)
MAINT	Maintenance Engineer	Command center	Maintenance, inventory, routes	Imran Khan (`maint`)

Role code	Name	Home screen	Modules	Demo user (ID)
OPERATOR	Shop-floor Operator	Production routes	Routes, manufacturing, kaizen	Mohan Lal (`operator`)

All roles except OPERATOR also see Home and Approvals. Every role can submit kaizen and apply for leave.

Permissions and the actions they unlock

Permission	Actions (UI `data-a` names)	Held by
indent.raise	newIndent, saveIndent, quickIndent	PLANT_HEAD, PURCHASE, STORES, QA, PROD, MAINT
po.create	newPOFromIndent, savePO, voiceVendor	PLANT_HEAD, PURCHASE
vendor.manage	newVendor, saveVendor, vendChecks	PURCHASE
cash.purchase	newCash, saveCash	PURCHASE, STORES
grn.post	newGRN, saveGRN, ocrDC	STORES
inv.issue / inv.transfer / inv.count	issueDo · newTransfer, xferDo · countStart, countSubmit	STORES; PLANT_HEAD, PROD (issue); PLANT_HEAD (count)
sales.order / sales.quote / sales.dispatch	newSO, saveSO, ocrPO, sched* · makeQuote · dispatch, dispatchDo	SALES
challan / scrap.sale	newRGP, saveRGP, rgpBack · newScrapSale, saveScrap	SALES, STORES · SALES, PLANT_HEAD
mfg.plan / mfg.entry	mrpIndents · saveProd, voiceEntry	PLANT_HEAD, PROD · + OPERATOR
route.move	batch*, metal*, mould*, lotNext, packDo	PLANT_HEAD, PROD, OPERATOR

Permission	Actions (UI `data-a` names)	Held by
qa.approve / ncr	qaDo, qaSave · newNCR, saveNCR, ncrNext	QA
mnt / mnt.breakdown	donePM, pmFromAI, closeBD · newBD, saveBD	MAINT, PLANT_HEAD · + PROD, OPERATOR
asset.manage / fin.voucher / fin.bill / fin.pay / gst	newAsset, runDep · newVoucher · ocrBill · payRun, payDo	ACCOUNTS
hr.manage / payroll	newEmp, candNext, screenCV · runPayroll	HR
kaizen.submit / kaizen.manage	newKaizen, saveKz · kzNext	all · PLANT_HEAD, HR
admin	user*, rule*, set*, resetDemo	ADMIN only
ai	AI copilot and agents	PLANT_HEAD, PURCHASE, PROD, SALES, ACCOUNTS

Access requirements

ID	Requirement	Priority
FR-ADM-01	Admin creates, edits, deactivates users and resets passwords; each user has one role and a plant scope	M
FR-ADM-02	Roles, modules and permissions are data, editable by admin; new custom roles can be cloned from a standard one	S
FR-ADM-03	A deactivated user is signed out on their next request	M
FR-ADM-04	Plant scope filters lists and reports ("All plants" or a list of units)	M
FR-ADM-05	Every create, approve, reject, setting change, sign-in and sign-out writes an audit row (time, user, role, action, reference)	M

ID	Requirement	Priority
FR-ADM-06	SSO with Microsoft Entra ID or Google Workspace, plus OTP fallback for shop-floor users without e-mail	S
FR-ADM-07	Shared shop-floor terminals use short PIN sign-in, auto sign-out after 10 minutes idle	S

V-Rub's real user list and role mapping come from sheet 2 (people), not yet returned.

4. Functional requirements by module

Each module below lists its screens (prototype tabs) and the requirements engineering must build. The prototype screen is the UX reference; where it shows simulated behaviour (OCR, voice, IoT), the requirement states the real integration.

4.1 Sales and dispatch (`sales`)

Screens: Sales orders · Customer schedules (Excel / e-mail) · RFQ & quotations · Dispatch & invoicing · Challans (RGP, FOC, job work) · Customers · Scrap & other sales · Sales & FG reports · Customer portal.

ID	Requirement	Priority
FR-SAL-01	Customer master with billing codes, GSTIN, payment terms, credit limit, part-number cross-reference	M
FR-SAL-02	Sales order with lines, rates from customer price list, delivery schedule per line	M
FR-SAL-03	Upload an OEM schedule (.xlsx/.csv) or paste an e-mail; preview rows matched to customer, PO and part; apply creates or updates SO schedule lines (`schedUpload`, `schedApply`)	M
FR-SAL-04	Mailbox watcher on the sales inbox parses schedule attachments automatically and queues them for preview	S

ID	Requirement	Priority
FR-SAL-05	Dispatch picks FG lots FIFO, blocks when FG stock is short, prints packing list and FG labels	M
FR-SAL-06	Tax invoice with IRN (e-invoice) and e-way bill via GSP; inter-unit invoice for transfers between GSTINs	M
FR-SAL-07	Returnable / non-returnable challans, job-work challans with due-back tracking (`rgpBack`)	M
FR-SAL-08	Scrap and old-asset sale routes through approval before invoicing	M
FR-SAL-09	Credit-limit and overdue check on SO and invoice: warn only until V-Rub decides	S
FR-SAL-10	Customer portal for schedules, dispatch status and invoices	C

4.2 Procurement (`procurement`)

Screens: Indents · Purchase orders · Supplier schedule · GRN & inspection · Returns · Cash purchase · Vendors · Reports.

ID	Requirement	Priority
FR-PRC-01	Indent types BOM, Consumable, Capital; capital indents carry description and value; route per approval matrix	M
FR-PRC-02	PO from one or more approved indents, or without indent when setting `requireIndent` is off	M
FR-PRC-03	PO approval by value band (incl. GST): up to ₹5 L, ₹5–25 L, above ₹25 L	M
FR-PRC-04	Vendor onboarding with GSTIN / PAN / bank checks; new vendors stay "Pending approval" and cannot receive POs	M
FR-PRC-05	Vendor rating on quality, delivery and price; MSME flag and 45-day payment tracking	M

ID	Requirement	Priority
FR-PRC-06	Supplier delivery schedule per PO line; late-line follow-up list	M
FR-PRC-07	GRN against PO, max receipt = pending qty + 10%; lot number, DC number, store; stock goes to QA hold	M
FR-PRC-08	Rejections create a purchase return and debit note	M
FR-PRC-09	Cash purchase with configurable per-bill limit (not enforced until V-Rub decides)	S
FR-PRC-10	RFQ to vendors, comparison sheet, minimum-quotes rule (configurable, off)	S

4.3 Inventory and stores (`inventory`)

Screens: Stock by store · Issue to production (FEFO) · Monthly stock count · Reorder (ROP / ROQ) · Shelf life & lots · Inventory ageing · Transfers.

ID	Requirement	Priority
FR-INV-01	Item master: code, type (RM / WIP / FG / spare), UoM, HSN, rate, shelf life (days), ROP, ROQ, category	M
FR-INV-02	Lot-level stock per store; every receipt creates a lot with received date and expiry = received + shelf life	M
FR-INV-03	Issue plans picks by earliest expiry, then oldest receipt; expired lots are blocked and listed; shortfall blocks the issue	M
FR-INV-04	No transaction may take any store/item/lot below zero	M
FR-INV-05	Valuation FIFO; ledger stores cost layers per receipt	M

ID	Requirement	Priority
FR-INV-06	Monthly count sheet per store; variances post only after approval (Plant Head → Accounts); rejected count goes to Recount	M
FR-INV-07	Inter-store and inter-plant transfers; inter-plant raises inter-unit invoice + e-way bill	M
FR-INV-08	Reorder suggestions below ROP with one-click indent	M
FR-INV-09	Code 39 lot labels at receipt; scan to issue, transfer and count	M
FR-INV-10	Ageing buckets and near-expiry alerts	S

4.4 Production routes (`routes`) and manufacturing (`manufacturing`)

Screens: Production routes board (Metal / Moulding / Extrusion lanes, compound batches) · Production plan · Shift production entry · WIP · Bills of material · MRP · Scrap · OEE & downtime.

ID	Requirement	Priority
FR-MFG-01	Compound master: recipe (phr per RM), maturation minimum and maximum hours, hardness spec	M
FR-MFG-02	Batch lifecycle: Weighed → Mixing → Mixed (maturing) → Awaiting QA → Ready → Consumed / Expired / Rejected	M
FR-MFG-03	Weighing records target vs actual per ingredient; RM issued FEFO; batch label printed	M
FR-MFG-04	Metal lots follow the METAL route with QA gates; RM consumed per part norm	M
FR-MFG-05	Moulding lot consumes compound by FIFO across several Ready batches and a metal lot that passed final metal QA	M

ID	Requirement	Priority
FR-MFG-06	Extrusion lot consumes EPDM compound FIFO; then joint moulding or flocking	M
FR-MFG-07	Each stage move (`lotNext`) records who, when, qty in/out, rejects; QA stages need a QA user	M
FR-MFG-08	Packing books FG stock with a lot label (`packDo`)	M
FR-MFG-09	Shift production entry (A/B/C) per machine: good qty, rejects by reason, downtime; alert if a shift ends without an entry	M
FR-MFG-10	BOM per FG with compound, metal insert and packing; MRP from SO schedules to indents	M
FR-MFG-11	OEE per press from shift entries	S
FR-MFG-12	Voice entry of production on the shop floor (English)	C

4.5 Quality (`quality`)

Screens: QA approvals (GRNs, batches, stage lots) · NCR & 8D · Incoming inspection · PPAP & APQP · SPC · Calibration.

ID	Requirement	Priority
FR-QA-01	One QA queue for GRN lots, compound batches and stage lots (`qaQueue`)	M
FR-QA-02	GRN accept / partial / reject; accepted qty moves from QA hold into store	M
FR-QA-03	Batch QA records hardness (and t90, SG); value outside the compound spec cannot be approved	M
FR-QA-04	NCR with 8D steps, linked to lot, supplier or customer complaint	M
FR-QA-05	Instrument calibration schedule with due alerts	S
FR-QA-06	PPAP document tracker per part and customer; SPC charts on key characteristics	C

4.6 Maintenance (`maintenance`)

Screens: Preventive maintenance · Breakdowns · Predictive (IoT) · Fixed asset register · Depreciation · Spares.

ID	Requirement	Priority
FR-MNT-01	Machine and mould master; PM schedules by days or shots; overdue list	M
FR-MNT-02	Breakdown ticket from any shop-floor user; close with cause, action, spares used; MTTR / MTBF	M
FR-MNT-03	Spares stock from Spares Store with reorder	M
FR-MNT-04	Fixed asset register and depreciation run (Companies Act and Income-tax)	S
FR-MNT-05	IoT condition monitoring	C (out of scope)

4.7 Finance and accounts (`finance`) and GST (`gst`)

Screens: Overview · General ledger · Payables · Receivables · Vouchers & expenses · Supplier & customer reports · Bank · Consolidation · Period lock; GST returns · GSTR-2B reconciliation · e-Invoice & e-way bill · TDS · Compliance calendar.

ID	Requirement	Priority
FR-FIN-01	Chart of accounts per unit; auto-postings from GRN, invoice, issue, production, payment	M
FR-FIN-02	Supplier bill booking with 3-way match (PO, GRN, invoice); price tolerance 0%; qty, rate or duplicate exceptions go to Accounts approval	M
FR-FIN-03	Payment run proposal by due date, MSME first; bank file export	M
FR-FIN-04	Receivables ageing and collections list	M
FR-FIN-05	Period lock: postings into a closed month refused; production postings allowed until day 1 of the next month (`periodOpen`)	M
FR-FIN-06	Consolidation across units with inter-unit elimination	S
FR-FIN-07	GSTR-1 / 3B data, GSTR-2B reconciliation, TDS	M
FR-FIN-08	Integration to existing accounting package if V-Rub keeps it as books of record	S

4.8 HR and kaizen (`hr`, `kaizen`)

ID	Requirement	Priority
FR-HR-01	Employee master, departments, reporting lines	M
FR-HR-02	Leave application with approval by department head; balances (demo default CL 7, SL 7, EL 15)	M
FR-HR-03	Attendance and shifts from biometric devices	S
FR-HR-04	Yearly appraisal: self → manager → HR approval	M

ID	Requirement	Priority
FR-HR-05	Kaizen submission by any user; dept head → Plant Head evaluation with score and saving	M
FR-HR-06	Payroll processing, recruitment	C (phase 2)

4.9 Approvals, AI, reports and administration

ID	Requirement	Priority
FR-APR-01	Approvals inbox "Waiting for me"; MD, Plant Head and Admin also see "All pending"	M
FR-APR-02	Bulk approve own items under ₹50,000	S
FR-AI-01	Copilot answers questions on live data within the user's permissions (e.g. "which batches expire today")	S
FR-AI-02	Agents: reorder suggestions, late-supplier chase, bill OCR, delivery-challan OCR, customer-PO OCR	S
FR-RPT-01	Report library with filters by unit and date, export to Excel/PDF	M
FR-ADM-08	Admin screens: Users & roles, Role access, Approval matrix, V-Rub business rules, Process flow, Audit log	M

5. Approval and workflow engine

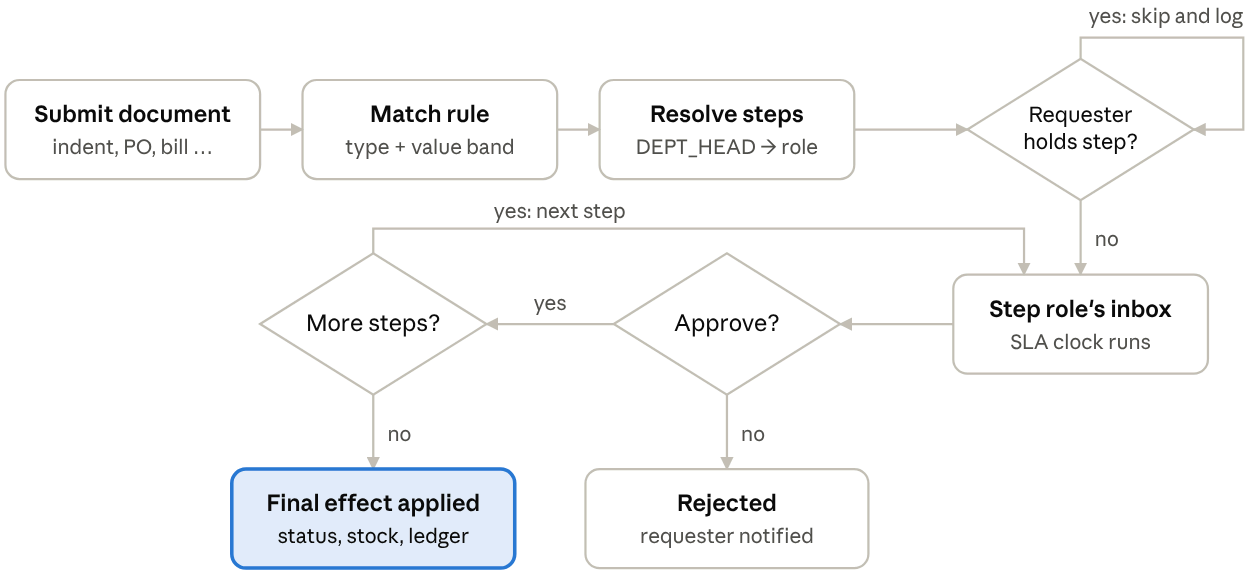
One generic engine routes eight document types. Rules are data (`S.rules` in the prototype, an `approval_rule` table in production), so V-Rub's admin changes routes without a release. A rule change applies only to documents submitted after it; in-flight documents keep the route they started with.

How a document is routed

1. **Match a rule** (`findRule`): first rule where `doc` matches, the document type is in `types` (or `types` is empty), and `min ⩽ amount ⩽ max`.

2. **Resolve steps** (`resolveSteps`): `DEPT_HEAD` becomes a role from the document's department via `DEPT_ROLE` (Production/Press/Mixing/Extrusion → PROD, Stores/Purchase → STORES, Maint → MAINT, Quality → QA, Sales → SALES, Accounts → ACCOUNTS, HR/Admin → HR, else PLANT_HEAD). Consecutive duplicate roles collapse into one step.
3. **Skip own role** (`startAppr`): leading steps held by the requester's role are logged as "Skipped · raised by this role". If no steps remain, the document is auto-approved.
4. **Queue**: the item appears in the inbox of every user whose role equals the current step (`buildApprovals` , flag `mine`).
5. **Decide** (`decide`): only the current-step role (or ADMIN, logged as override) may act. Approve moves to the next step or finalises; reject finalises as rejected. The document's `final(d, ok)` hook applies the business effect (e.g. count variance posts stock; bill on hold becomes approved with debit note).
6. **SLA**: age per step vs rule SLA; past-SLA items are flagged and escalated.

A document waits only for roles other than the requester



approval routing · one engine, eight document types

If every step is held by the requester's own role, the document is auto-approved; otherwise it loops through the inbox until the last step or a rejection.

Document types (`DOCS`)

Type	Collection	Pending when	Amount used for banding	Department source	Final effect
indent	indents	status = Pending approval	Capital: capVal; else qty × item rate	indent.dept	Approved / Rejected
po	pos	appr = Pending approval	PO value × 1.18	Purchase	appr Approved, status Open
bill	bills	On hold or Blocked	taxable + GST	Accounts	Approved (debit note if on hold)
scrap	scrapSales	Pending approval	amount	Production	Invoiced
count	counts	Pending approval	abs(variance)	Stores	Posted (stock adjusted) or Recount
leave	leaves	Pending	—	employee dept	Approved / Rejected
kaizen	kaizens	Submitted / Under evaluation	saving	area	Approved; step 2 sets Under evaluation
review	reviews	Manager review / HOD approval	—	employee dept	Closed

Default rules (HeyyBot proposal, pending V-Rub sheet 3)

Rule ID	Document	Band / types	Steps	SLA per level
R-IND	Indent	BOM, Consumable	Dept head → Plant Head	24 h
R-IND-CAP	Indent	Capital	Dept head → Plant Head → MD	72 h
R-PO-1	PO	up to ₹5 L	Purchase	24 h
R-PO-2	PO	₹5 L – ₹25 L	Purchase → Plant Head	24 h
R-PO-3	PO	above ₹25 L	Purchase → Plant Head → MD	24 h
R-BILL	Supplier invoice exception	any	Accounts	48 h

Rule ID	Document	Band / types	Steps	SLA per level
R-SCRAP	Scrap / asset sale	any	Plant Head → Accounts	48 h
R-COUNT	Stock count variance	any	Plant Head → Accounts	48 h
R-LEAVE	Leave	any	Dept head	24 h
R-KZ	Kaizen	any	Dept head → Plant Head	168 h
R-REV	Annual review	any	Dept head → HR	120 h

Reminders and escalation (proposal, pending V-Rub sheet 6)

When	Action	Channel
50% of SLA	Reminder to approver	In-app, e-mail
90% of SLA	Second reminder with one-tap approve	WhatsApp
SLA breached	Escalate one level, notify requester	WhatsApp, e-mail
Shift ends without production entry	Alert line supervisor	In-app, WhatsApp

Engine requirements

ID	Requirement	Priority
FR-WF-01	Decide is atomic and idempotent: a second decision on the same step returns "Already decided" (optimistic lock on `ap.at` + version)	M
FR-WF-02	Each step logs role, user, acting-as role, decision, note, timestamp, override flag	M

ID	Requirement	Priority
FR-WF-03	Delegation: a user can delegate their role's approvals to another user for a date range	S
FR-WF-04	Approve / reject from e-mail or WhatsApp via signed one-time link	S
FR-WF-05	New document types are added by registering a type adapter (pending, amount, dept, title, final) — no engine change	M

6. V-Rub business rules and production routes

V-Rub answered 26 business-rule questions in sheet 7 of the workflow template: 14 are applied, 3 are configurable but awaiting a decision, and 9 are unanswered. Every rule is held in a `settings` record, so changing an answer is a setting, not a code change.

6.1 Business rules

ID	Area	Rule	V-Rub answer	System behaviour	Setting key	Status
RULE-01	Procurement	Excess receipt over PO qty	Yes, 10%	GRN refuses more than pending + 10%	grnTolPct = 10	Applied
RULE-02	Procurement	PO vs invoice price difference	No	0% tolerance; any rate variance holds the bill for Accounts	priceTolPct = 0	Applied
RULE-03	Procurement	PO without approved indent	Yes initially; build indent-first release	Toggle: off = allowed now, on = indent required	requireIndent = false	Applied
RULE-04	Procurement	Cash purchase limit / petty cash	Later	No limit enforced	cashLimit = null	Awaiting
RULE-05	Procurement	Minimum quotes before PO	Later	Not enforced	minQuotes = null	Awaiting

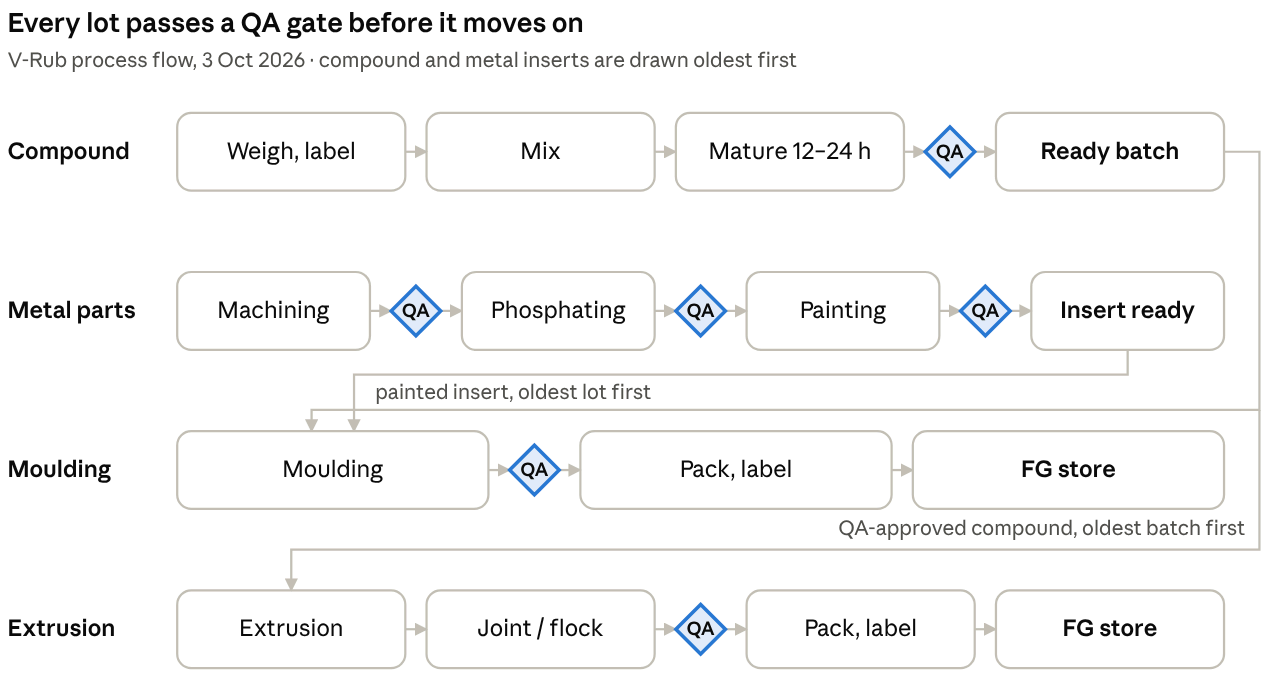
ID	Area	Rule	V-Rub answer	System behaviour	Setting key	Status
RULE-06	Stores	Valuation	FIFO	FIFO cost layers	valuation = FIFO	Applied
RULE-07	Stores	Negative stock	No	Issue, transfer, dispatch, moulding block when short	negativeStock = false	Applied
RULE-08	Stores	Lot issue rule	First expiry first out	FEFO picking; expired lots blocked	issueRule = FEFO	Applied
RULE-09	Stores	Physical count	Monthly	Count sheet per store; variance after approval	countFreq = Monthly	Applied
RULE-10	Sales	Margin needing MD approval	Later	Not enforced	marginMD = null	Awaiting
RULE-11	Sales	OEM schedules	Excel / e-mail	Schedule import	schedFormat	Applied
RULE-12	Production	Reporting	Per shift	Shift entry A/B/C; missed-entry alert	prodReporting = Per shift	Applied
RULE-13	Production	Backflush or actual issue	Actual issue	No backflush; FEFO issue against batch / WO	consumption = Actual issue	Applied
RULE-14	Finance	Separate GSTINs, inter-unit invoices	Yes	Inter-plant transfer raises inter-unit invoice + e-way bill	interUnit = true	Applied
RULE-15	Finance	Period lock	Last day of month; production open to day 1 of next month	Posting refused in closed periods	lockDay, prodLockExtraDays = 1	Applied
RULE-16	HR	Appraisal cycle	Yearly	FY 2026-27 cycle	appraisal = Yearly	Applied
RULE-17	General	Language	English	English UI and messages	language = English	Applied

Unanswered (defaults in use): vendor payment terms (MSME 45-day tracking on); customer payment terms by type; credit-limit block (warn only); rejection % target per line; who may post into a closed period (nobody); banks and host-to-host (file export); shifts, leave, overtime, payroll dates, attendance devices (demo defaults); who maintains masters (admin and module owners); document numbering (`PO-26-0861` style).

6.2 Production routes (from V-Rub process-flow PDF, 3 Oct 2026)

Route	Stages in order (QA gates in bold)	Output
METAL — Metal parts	Machining → **QA check** → Phosphating, barcoding → **QA check & issue to painting** → Painting → **QA check & issue to moulding** → Ready for moulding	Metal insert lot for moulding
MOULD — Rubber moulded parts	Moulding → **QA check** → Packing, barcoding → FG store	FG lot
EXTR — Extruded & flocked parts	Extrusion → Joint moulding or flocking → **QA check** → Packing, barcoding → FG store	FG lot

Upstream of all three: vendor approval → PO & schedule → RM receipt → store, barcoding, QA approval → issue for production (FIFO & expiry check). Rubber lanes add batch weighing & barcoding → mixing → maturation & QA approval.



Moulding cannot start until a matured, QA-approved compound batch and a painted insert lot are both available.

6.3 Compound batches

Compound	Description	Lane	Min maturation	Max age	Hardness spec (Shore A)
VR-C60	NR compound, 60 Shore A	Rubber	24 h	120 h	55–65
VR-C70	NR/SBR compound, 70 Shore A	Rubber	24 h	120 h	65–75
VR-E65	EPDM extrusion compound, 65 Shore A	Extrusion	12 h	96 h	60–70

Batch status is derived, never stored as truth (`bStatus`): Weighed / Mixing → **Maturing** while age < min → **Awaiting QA** → **Ready** → **Consumed** when kg left = 0, **Expired** when age > max, **Rejected** on failed QA. Production must compute it server-side from `mixed_at` , QA result and remaining kg.

6.4 Consumption rules at moulding and extrusion

- Compound is allocated across Ready batches oldest-first until the need is met (`fifoAlloc`); one lot may draw from several batches, and each draw is recorded for traceability.
- A moulded part with a metal insert must draw from a metal lot at "Ready for moulding", oldest first.
- If total Ready compound or metal is short, the moulding lot cannot be created.
- QA stages can only be passed by a user with `qa.approve`; non-QA stages need `route.move`.
- Every lot carries a Code 39 label (`labelHTML`): code, part, qty, route, stage, dates; batch labels add "usable from" and "use before".

7. Prototype code walkthrough

The prototype is a single self-contained HTML file (~4,200 lines, no framework, no build step beyond concatenation). It is the behavioural reference for the production build: port its rules, not its architecture. Everything in it runs in the browser; production moves state, rules and permission checks to the server.

7.1 Files and build

File	Lines	Contents
shell.html	303	Layout (rail, top bar, view, drawer, copilot, palette, login overlay), all CSS with light/dark tokens, `/*__JS__*/` placeholder
data.js	285	`seed()`: demo master and transaction data; `BUS` (units), `STORES`, date helpers `dAgo`, `dFwd`
core.js	254	State load/save, `ACT` action registry and click dispatcher, `table()`, `openDrawer()`, router, `NAV`, command palette, CSV export, AI copilot
pages1.js	223	Home, sales, procurement pages
pages2.js	146	Inventory, manufacturing, quality, maintenance pages

File	Lines	Contents
pages3.js	160	Finance, GST, HR, kaizen, integrations, reports, go-live, benchmark pages
vrub.js	862	V-Rub layer: roles, permissions, login, approval engine, sync, FEFO, barcodes, production routes, compound batches, QA queue, schedule import, period lock, admin, business rules

```
cat data.js core.js pages1.js pages2.js pages3.js vrub.js > all.js
node --check all.js            # syntax gate
# inject all.js into shell.html at /*__JS__*/  ->  heyyerp-vrub.html
```

Test harness (Playwright, Chromium): `t2.py` opens every screen for every role (about 200 screens, zero console errors); `t3.py` runs a multi-tab hand-off (stores raises indent → plant head approves → purchase creates PO → stores posts GRN → QA accepts → stock in store); `t4.py` clicks every action button on every screen.

7.2 Runtime model

- **Routing**: hash routes `#/<module>/<tab>/<arg>`. `route()` checks sign-in and `modAllowed(module)`, then renders `PAGES[module](tab, arg)` into `#view` and calls `afterRender`.
- **Actions**: buttons carry `data-a="<action>" data-p="<param>"` (helper `A(fn, p)`). One document-level click listener looks up `ACT[action]`, checks `actAllowed(action)` against `ACTPERM`, then calls it. Pages extend earlier pages by wrapping (`const _home = PAGES.home; PAGES.home = …`).
- **State**: one object `S`, persisted to `localStorage['heyyerp-vrub-v4']` by `save()`. `S.meta.v !== 4` triggers a reseed (`fullSeed()`).
- **Session**: `sessionStorage['heyyerp-vrub-session']` holds the user ID, so each browser tab can be a different user; "keep me signed in" also writes `localStorage`. Passwords are djb2 hashes (`hashPw`) — demo only.
- **Multi-user sync**: with the artifact `db` capability, each top-level key of `S` is one document `erp/<key>` with body `{v, by, at}`. `syncPush()` diffs each key against `lastSent` and writes changed keys; `onSnapshot` applies remote keys unless a local

write for that key is pending; a re-render is deferred while a drawer or input is active. Without `db`, the browser `storage` event syncs tabs on one device. `room` gives presence (who is signed in).

- **AI**: `sample` capability; `copContext()` builds a text snapshot of below-ROP items, receivables, payables, open SOs and WOs for the copilot prompt.

7.3 State schema (`S`)

Key	Holds	Key fields
meta	version	v = 4, seededAt
users	accounts	id, name, role, dept, pw (hash), active, plants
settings	V-Rub business rules	grnTolPct, priceTolPct, requireIndent, valuation, negativeStock, issueRule, countFreq, prodLockExtraDays, …
rules	approval rules	id, doc, name, types[], min, max, steps[], sla
audit	audit log (last 400)	at, user, role, action, ref
items	item master + stock	code, name, type, uom, rate, hsn, stock{store: qty}, rop, roq, shelf, cat
lots	lot stock	item, lot, qty, rcv, store
compounds	compound master	code, name, lane, matureH, maxH, hard[min,max], recipe[[item, phr]]
batches	compound batches	no, compound, kg, kgLeft, stage, mixedAt, qa, hardness, lots[]
wip	route lots	id, route, item, qty, stage (index), createdAt, hist[], src[] (batch / metal draws)
vendors, customers, custPrice	masters	code, name, gstin, terms, msme, rating q/d/p, status

Key	Holds	Key fields
indents, pos, grns, bills	procure-to-pay	no, status/appr, lines, recv[], qc, match, `ap` (approval state)
sos, schedules, invoices, challans, scrapSales	order-to-cash	no, cust, lines, due, irn, ewb, so link
issues, counts	stores transactions	no, lines, variance, status
wos, prodLog, boms	manufacturing	no, item, qty, done, rej, line, shift
machines, pms, breakdowns, assets, calib	maintenance & quality	code, status, due
ncrs, ppap	quality	no, step, item
emps, leaves, reviews, kaizens, candidates, depts	HR	id, dept, status, `ap`
vouchers, voiceCalls, rfqs, quotes	misc	—
seq	document counters	ind, po, grn, so, inv, cb, ml, mo, ex, …

Every approvable document carries `ap = {rule, ruleName, steps[], at, sla, startedAt, stepAt, log[]}`.

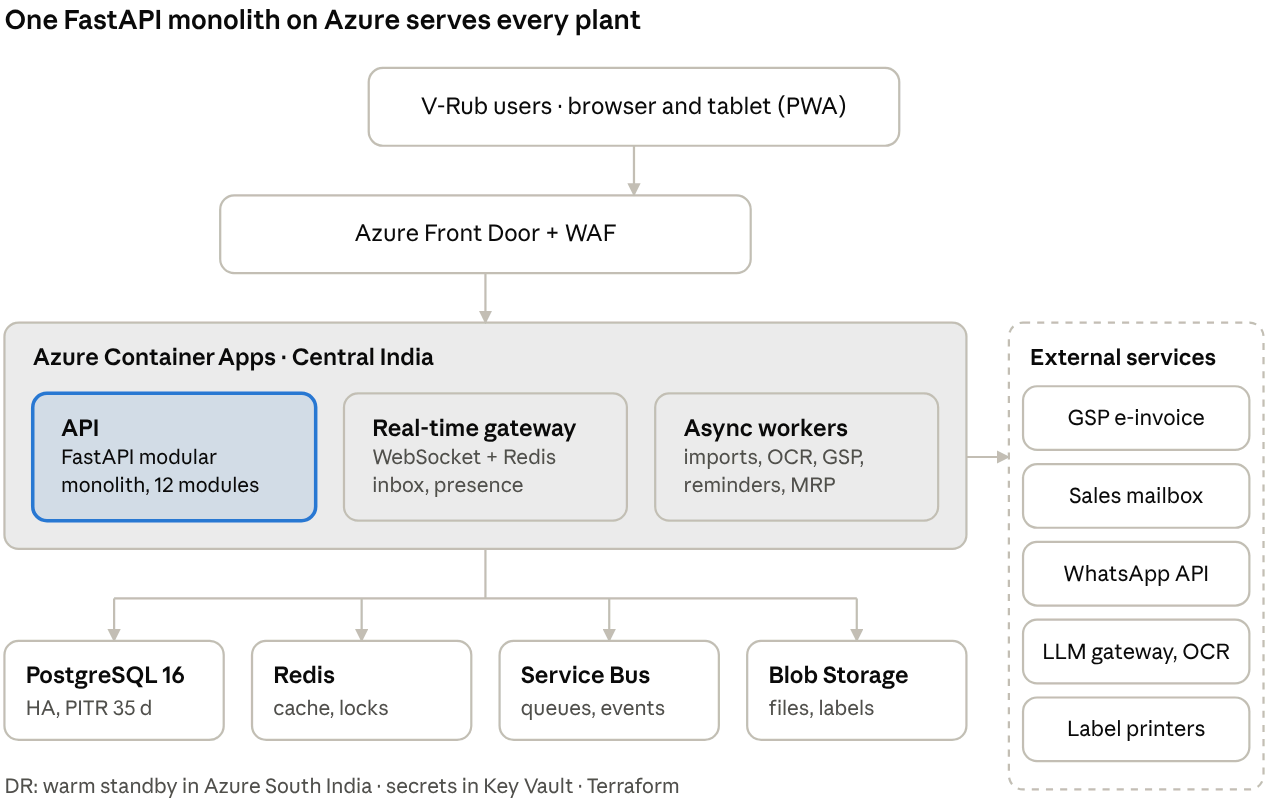
7.4 Key functions to port

Function	File	What it does	Production home
hasPerm, actAllowed, modAllowed	vrub.js	Role → permission / module checks	Auth middleware + policy service
findRule, resolveSteps, deptRole	vrub.js	Rule match and step resolution	Workflow service
startAppr, attachApprovals, decide	vrub.js	Start, seed and decide approvals	Workflow service (transactional)

Function	File	What it does	Production home
buildApprovals	vrub.js	Inbox query with SLA age	`GET /approvals?mine=true`
fefoPlan, postIssue, lotExpiry	vrub.js	FEFO pick plan, stock decrement, expiry	Inventory service
fifoAlloc	vrub.js	Oldest-first allocation across batches / lots	Shared domain library
bStatus, bTimer	vrub.js	Derived compound-batch status	Production service (computed column / view)
qaQueue	vrub.js	GRNs + batches + stage lots awaiting QA	Quality service query
periodOpen	vrub.js	Period-lock check incl. production grace day	Finance service guard on every posting
barcode, labelHTML	vrub.js	Code 39 SVG and label layout	Label service (ZPL for Zebra printers)
schedUpload, schedApply	vrub.js	Parse Excel (SheetJS 0.18.5) and apply to SOs	Sales service + import worker
syncPush, applyRemote	vrub.js	Key-level sync	Replaced by REST + WebSocket events
audit, stamp	vrub.js	Audit row and who/role stamp	Audit middleware

8. Target architecture, data model and APIs

Production heyyERP is a modular monolith: Python 3.12 / FastAPI, PostgreSQL 16, React 18 + TypeScript, on Azure Central India — the same stack as the PRD. It is multi-tenant from day one (`tenant_id` on every row, Postgres row-level security), so V-Rub is tenant 1 and the next auto-component customer needs no rewrite. Modules are Python packages with their own schema and service layer; they talk through in-process service interfaces and domain events, never by reading each other's tables.



target architecture · Azure Central India

All user traffic enters through Front Door to one API; slow or external work goes to workers through Service Bus.

8.1 Layers

Layer	Technology	Responsibility
Web app	React 18, TypeScript, Vite, TanStack Query, React Router, design tokens from prototype CSS	All screens; role-aware nav; offline-tolerant shop-floor views (PWA)
Edge	Azure Front Door + WAF, CDN	TLS, WAF rules, static assets, geo restriction to India if required
API	FastAPI (async), Pydantic v2, SQLAlchemy 2, Alembic	REST `/api/v1`, OpenAPI spec, auth middleware, permission guard, audit middleware

Layer	Technology	Responsibility
Real-time	WebSocket gateway (FastAPI + Redis pub/sub)	Approval inbox counts, "updated by" toasts, presence — replaces prototype `db` / `room` sync
Domain modules	sales, procurement, inventory, production, quality, maintenance, finance, gst, hr, workflow, admin, ai	Business rules ported from prototype (Section 7.4)
Async workers	Celery or Dramatiq on Azure Service Bus; scheduler	Schedule import, OCR, GSP calls, reminders/escalations, MRP nightly, report export
Data	Azure Database for PostgreSQL Flexible Server 16 (zone-redundant HA), Azure Cache for Redis, Blob Storage	Transactions, cache/sessions/locks, documents and labels
AI	LLM gateway (Azure OpenAI + Anthropic Claude), HeyyBot Smart OCR on Azure AI Document Intelligence	Copilot, agents, bill / DC / customer-PO OCR
Integrations	GSP (e-invoice, e-way bill), SMTP/Graph mailbox, WhatsApp Business API, bank file, Tally/accounting export, biometric devices, Zebra printers	Adapters behind one integration module with retry and dead-letter

8.2 Core data model (PostgreSQL)

All tables carry `tenant_id`, `unit_id` (where plant-specific), `created_at`, `created_by`, `updated_at`, `version` (optimistic lock). Money in `numeric(18,2)`, quantities in `numeric(18,3)`.

Domain	Tables (key columns)
Org & security	tenant, unit (gstin, state_code), store, department, app_user (role_id, unit_scope[], status), role, permission, role_permission, role_module, session, audit_log (append-only)
Settings	business_setting (key, value jsonb, effective_from) — holds grnTolPct, priceTolPct, requireIndent, etc.
Workflow	approval_rule (doc_type, types[], min_amt, max_amt, sla_hours), approval_rule_step (seq, role_code or DEPT_HEAD), approval_instance (doc_type, doc_id, rule_id, current_step, status, started_at, step_at), approval_action (step, role, user, acting_as, decision, note, override, at), delegation
Items & stock	item (type, uom, hsn, shelf_days, rop, roq), lot (item, lot_no, received_on, expiry_on, store, status: QA_HOLD / AVAILABLE / QUARANTINE / BLOCKED), stock_ledger (item, lot, store, qty ±, ref_type, ref_id, cost_layer_id), cost_layer (FIFO: item, receipt_ref, qty_left, unit_cost), stock_count, stock_count_line
Procure-to-pay	vendor (status, msme, rating), indent, indent_line, purchase_order, po_line, po_schedule, grn, grn_line, purchase_return, supplier_bill, bill_match (po/grn/bill variance)
Order-to-cash	customer, price_list, sales_order, so_line, so_schedule (source: excel / email / manual, import_id), schedule_import (file, parsed_rows jsonb, status), dispatch, invoice (irn, ack_no, ewb_no), inter_unit_invoice, challan

Domain	Tables (key columns)
Production	compound (mature_h, max_h, hardness_min/max), compound_recipe_line, compound_batch (mixed_at, qa_status, hardness, kg, kg_left), batch_weighing_line, route, route_stage (seq, is_qa_gate, is_end), prod_lot (route, item, qty, stage_seq), prod_lot_move (from, to, qty_in, qty_out, rejects, user), lot_consumption (prod_lot, source_type batch/metal_lot/rm_lot, source_id, qty), shift_production, bom, bom_line, work_order
Quality	qa_inspection (subject_type grn_lot / batch / prod_lot, result, params jsonb), ncr, ncr_step, calibration
Maintenance	machine, mould, pm_plan, pm_task, breakdown, spare_issue, fixed_asset, depreciation_run
Finance & GST	account, journal, journal_line, period (unit, month, status, prod_grace_days), payment, payment_run, gst_return_data, tds_entry
HR	employee, leave_type, leave_request, leave_balance, attendance, appraisal, kaizen

Batch status stays derived: a view `v_compound_batch_status` computes Maturing / Awaiting QA / Ready / Expired / Consumed from `mixed_at`, `now()`, QA result and `kg_left`.

8.3 API design

REST, JSON, `/api/v1/<module>/<resource>`; cursor pagination; `If-Match` with `version` on updates; idempotency key header on all POSTs that create stock or money movements.

Endpoint	Purpose	Permission
`POST /auth/login`, `/auth/sso/callback`, `/auth/refresh`	Sign-in; JWT access (15 min) + refresh (httpOnly cookie)	—

Endpoint	Purpose	Permission
`GET /me`	User, role, modules, permissions, unit scope	any
`GET /approvals?mine=true` / `?all=true`	Inbox	any / MD, PLANT_HEAD, ADMIN
`POST /approvals/{doc_type}/{doc_id}/decide {decision, note}`	Approve / reject current step	current-step role or ADMIN
`POST /procurement/indents`, `/purchase-orders`, `/grns`	Create; submit starts approval	indent.raise, po.create, grn.post
`POST /inventory/issues/plan` → `POST /inventory/issues`	FEFO plan preview, then post	inv.issue
`POST /inventory/counts/{id}/submit`	Submit count; starts approval	inv.count
`POST /production/batches`, `/batches/{no}/mix-start`, `/mix-done`	Compound batch lifecycle	route.move
`POST /production/lots`	Create metal / moulding / extrusion lot with FIFO allocation	route.move
`POST /production/lots/{id}/next`	Move to next stage (QA gate needs qa.approve)	route.move / qa.approve
`GET /quality/queue`, `POST /quality/inspections`	QA queue and decisions	qa.approve
`POST /sales/schedule-imports` (multipart) → `POST /{id}/apply`	Excel/e-mail schedule import	sales.order
`POST /sales/dispatches`, `/invoices/{no}/irn`	Dispatch, e-invoice	sales.dispatch
`GET/PUT /admin/approval-rules`, `/admin/settings`, `/admin/users`	Admin configuration	admin

Endpoint	Purpose	Permission
`GET /labels/{type}/{id}?format=zpl|pdf`	Code 39 label	per subject
`POST /ai/ask`	Copilot query, scoped to caller's permissions	ai

8.4 Domain events (outbox pattern)

Writes append to an `outbox` table in the same transaction; a relay publishes to Service Bus topics. Consumers: notifications, real-time gateway, finance auto-posting, search index, AI context cache.

Event	Emitted by	Consumed by
approval.step_pending / decided	workflow	notifications (in-app, e-mail, WhatsApp), real-time inbox
grn.posted, qa.decided	procurement, quality	inventory (QA hold → available), finance (GRN-IR accrual)
stock.issued, lot.moved, lot.packed	inventory, production	finance (WIP / FG valuation), traceability index
invoice.created	sales	GST adapter (IRN, e-way bill), finance (AR)
period.closed	finance	all posting services (guard cache refresh)

8.5 Integrations

System	Method	Phase
GSP for e-invoice / e-way bill	REST via licensed GSP (charges by V-Rub)	1
Sales mailbox	Microsoft Graph / IMAP poller; attachments to schedule import	1
WhatsApp Business API	Template messages for approvals and alerts	1
Barcode printers	ZPL over network print server at each plant	1

System	Method	Phase
Accounting package (if retained)	Daily voucher export (XML/CSV)	1, if needed
Bank	Payment file export; host-to-host later	1 / 2
Biometric attendance	Device vendor API or CSV pull	2
Machines / PLCs, OEM portals	Out of scope	—

9. Infrastructure, environments, CI/CD, security and NFRs

Everything runs in Azure Central India inside one VNet with private endpoints; DR is a warm standby in Azure South India. All infrastructure is Terraform; nothing is created by hand in the portal.

9.1 Azure resources (production)

Resource	SKU / sizing at go-live (≈ 25–50 users)	Notes
Azure Container Apps environment	api: 2–6 replicas (1 vCPU / 2 GiB); worker: 1–4; ws-gateway: 2	KEDA scaling on HTTP and queue depth
PostgreSQL Flexible Server 16	General Purpose D4ds_v5, 256 GB, zone-redundant HA	PITR 35 days; read replica for reports from phase 2
Azure Cache for Redis	Standard C1	Sessions, rate limits, locks, pub/sub
Service Bus	Standard	Queues: imports, ocr, gsp, notify; topics for domain events
Blob Storage	Hot, ZRS; lifecycle to Cool after 90 days	Attachments, labels, exports; ClamAV scan on upload
Front Door Standard + WAF	OWASP rule set	Custom domain e.g. `vrub.heyyerp.com`

Resource	SKU / sizing at go-live (≈ 25–50 users)	Notes
Key Vault	Standard	All secrets; managed identities only
Container Registry	Basic	Signed images
Monitor, Application Insights, Log Analytics	30-day retention, 1-year archive	OpenTelemetry traces, metrics, logs

9.2 Environments

Env	Purpose	Data	Deploy trigger
dev	Developer integration	Seed data (prototype `fullSeed` as fixtures)	Every merge to `main`
qa	Automated + manual testing	Seed + generated volume	Nightly / on demand
uat	V-Rub key users	Migrated V-Rub masters (masked where needed)	Release candidate tag
prod	Live	Live	Manual approval after UAT sign-off
dr	Warm standby (South India)	Geo-replica + geo-redundant backups	Terraform; failover runbook

9.3 Repository and CI/CD

Monorepo: `apps/web` (React), `apps/api` (FastAPI, one package per module), `apps/worker`, `packages/domain` (pure rules: FEFO, FIFO allocation, batch status, rule matching — ported from `vrub.js` with unit tests), `infra/terraform`, `e2e/` (Playwright, ported from `t2.py`–`t4.py`).

1. Pull request: lint (ruff, eslint), type check (mypy, tsc), unit tests, Alembic migration check, SAST (CodeQL / Semgrep), dependency and container scan (Trivy).
2. Merge to main: build and sign images, push to ACR, deploy to dev, run smoke and API contract tests.

3. Release tag: deploy to qa, then uat; run full Playwright suite including role × screen and multi-user hand-off tests.
4. Production: manual approval, blue/green revision swap in Container Apps, DB migrations expand-then-contract, automatic rollback on failed health checks.

GitHub Actions with OIDC federation to Azure; no long-lived cloud secrets in CI.

9.4 Security

Area	Requirement
Identity	Entra ID / Google SSO; local accounts with Argon2id hashes, lockout after 5 failures; OTP or PIN for shop floor; MFA for ADMIN, MD, ACCOUNTS
Authorisation	Every endpoint declares a permission; policy check server-side; unit scope applied in queries; Postgres RLS on `tenant_id`
Data	TLS 1.2+ everywhere; encryption at rest (platform keys, CMK option); PII fields (bank, PAN) encrypted at column level
Audit	Append-only `audit_log` for sign-in, create, approve, reject, setting and rule changes, exports; retained 8 years (GST / Companies Act)
App security	OWASP ASVS L2; CSP, CSRF protection, input validation via Pydantic; file-type and AV checks on uploads
AI	Copilot queries run through the same permission layer; no tenant data used to train models; prompt and response logging with PII redaction
Compliance	DPDP Act 2023 obligations; data resident in India

9.5 Non-functional requirements

ID	Requirement	Target
NFR-01	Availability (monthly, excl. planned maintenance)	99.5%
NFR-02	API latency, p95 for reads / writes	⩽ 300 ms / ⩽ 600 ms
NFR-03	Page interactive on a mid-range Android tablet over 4G	⩽ 3 s
NFR-04	Concurrent users without degradation	150 (3× expected peak)
NFR-05	Approval event to approver's inbox	⩽ 5 s
NFR-06	Schedule import of 2,000 rows to preview	⩽ 15 s
NFR-07	RPO / RTO	15 min / 4 h
NFR-08	Backups	Daily full + PITR 35 days; quarterly restore test
NFR-09	Browser support	Latest Chrome, Edge, Safari; Android 10+ tablets
NFR-10	Accessibility	WCAG 2.1 AA on all forms and tables
NFR-11	Localisation	English UI; Indian number format (lakh / crore) and INR
NFR-12	Data consistency	No negative stock and no double approval under concurrent requests (row locks / `SELECT … FOR UPDATE` on lot and approval_instance)

10. Migration, testing, rollout, open items and sign-off

Go-live is planned 14 weeks after kick-off, followed by 30 days of hyper-care, matching quotation HYB-2026-71.

10.1 Rollout plan

Phase	Weeks	Exit gate
1. Discovery & BRD	1–3	BRD v1.0 signed; V-Rub sheets 1–6 and 8 received; approval matrix frozen
2. Build & configure	4–9	All M requirements demo-able in qa; first data load done
3. UAT & training	10–13	UAT scripts passed with zero open Sev-1/Sev-2; key users trained; final data load reconciled
4. Go-live & hyper-care	14 + 30 days	Cut-over checklist complete; first month-end closed in heyyERP

Cut-over is at a month start so opening balances match a closed month in the old books. Parallel run for one week on stores and dispatch.

10.2 Data migration

Object	Source	Template owner	Validation
Units, stores, GSTINs, departments	Sheet 1	V-Rub	Admin review
Users and roles	Sheet 2	V-Rub	Each user signs in during UAT
Items, HSN, UoM, shelf life, ROP/ROQ	Current item list	V-Rub stores	No duplicate codes; HSN valid
Compounds and recipes, BOMs, routes	Process sheets	V-Rub production / QA	Sample FG costed and approved
Vendors, customers, price lists	Accounting package	V-Rub accounts	GSTIN and PAN validated

Object	Source	Template owner	Validation
Open POs, open SOs and schedules	Spreadsheets	Purchase, sales	Line count and value tie-out
Opening stock by lot and store	Physical count at cut-over	Stores	Count = ledger; value = books
Opening AR / AP, trial balance	Accounting package	Accounts	TB ties to the last closed month

Loads run through the same API validations as user entry (no direct SQL inserts), with a dry-run report before each load.

10.3 Testing

Level	Scope	Tooling	Owner
Unit	Domain rules: FEFO, fifoAlloc, batch status, rule matching, period lock, tolerances	pytest, Hypothesis property tests	Engineering
API / contract	Every endpoint incl. 403 for every role without the permission	pytest + schemathesis on OpenAPI	Engineering
End-to-end	Role × screen sweep; multi-user hand-offs (indent → PO → GRN → QA → store; batch → mature → QA → mould → pack → dispatch)	Playwright	QA
Concurrency	Two users approving the same step; two issues on the last lot	Locust + targeted race tests	Engineering
Performance	NFR-02 to NFR-06 at 150 users	k6 / Locust	Engineering
Security	OWASP ZAP, dependency and container scans, pen test before go-live	ZAP, Trivy, external tester	HeyyBot
UAT	Business scenarios per module, signed by key users	UAT scripts in the doc library	V-Rub

10.4 Open items and decisions needed

- [ ] V-Rub to return workflow template sheets 1–6 and 8 (company setup, people, approval matrix, process list, as-is flows, alerts, sign-off).
- [ ] Replace default approval matrix (Section 5) with V-Rub's sheet 3.
- [ ] Decide cash purchase limit, minimum quotes before PO, and quote margin needing MD approval.
- [ ] Answer the 9 open business-rule questions in Section 6.1.
- [ ] Confirm real GSTINs, plant names and store list (prototype values are placeholders).
- [ ] Confirm whether the existing accounting package stays as books of record (drives FR-FIN-08).
- [ ] Select GSP for e-invoice / e-way bill; V-Rub to hold the GSP contract.
- [ ] Confirm number of login accounts at go-live (billing basis).
- [ ] Reconcile the 14-week client implementation plan with the PRD's 28-week product build plan: confirm which platform modules are ready before V-Rub kick-off.

10.5 Sign-off

Party	Name	Role	Date	Signature
V-Rub Suspensions Pvt. Ltd.		Managing Director		
V-Rub Suspensions Pvt. Ltd.		Plant Head		
HeyyBot		CBO		
HeyyBot		Engineering lead		